

Reproducibility guide

Repository source: REPRODUCIBILITY.md

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This guide explains how to reproduce the symbolic checks, numerical cross-checks, manuscript build, and typeset Markdown documentation from a fresh clone.

Continuous-integration environments

The GitHub Actions workflow uses:

- Ubuntu 24.04;
- Python 3.13;
- the exact Python package versions in `requirements.txt`;
- R 4.6.1, using base R only;
- Pandoc 3.10.1;
- XeLaTeX, `latexmk`, and the TeX packages installed by the workflow; and
- `pdfinfo`, `pdffonts`, and `pdftotext` from Poppler for PDF preflight.

The workflow file is `.github/workflows/verification.yml`.

Recorded successful clean clone

A clean-clone reproduction was completed on 2026-09-04 for commit `96cea57275070f21babf30477ab1b128ec0f5eb6`. The isolated clone ran the Python and R suites, rebuilt the manuscript and every Markdown-derived PDF, passed the metadata and local checksum checks, and remained byte-clean afterward. The full Poppler PDF preflight was delegated to the three-job GitHub Actions run, which was green for the same commit.

The complete environment and interpretation are recorded in [CLEAN_CLONE_CHECK.md](#).

Fresh-clone procedure

Clone the repository and enter its root directory:

```
git clone YOUR_REPOSITORY_URL
cd minvol-degree3-spectral-bound
```

Replace `YOUR_REPOSITORY_URL` with the repository's actual clone URL.

1. Python checks

Create and activate a virtual environment if desired, then install the pinned requirements and run the suite:

```
python3 -m venv .venv
. .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
./run_python_checks.sh
```

A successful run ends with all of the following:

```
EXACT DEGREE-3 IDENTITY CHECK: PASS
ALTERNATIVE COMPACT-FORM CHECK: PASS
PAIRING-COUNT CHECK: PASS
BOUND ARITHMETIC CHECK: PASS
NUMERICAL STRESS TESTS: PASS
```

2. R checks

The R scripts use base R only:

```
./run_r_checks.sh
```

A successful run contains:

```
R PAIRING-COUNT CHECK: PASS
R NUMERICAL STRESS TESTS: PASS
```

The runner records both outputs and the R session information in `expected/`.

3. Manuscript build

With Python 3, `latexmk`, and a standard LaTeX installation available:

```
./build_paper.sh
```

The resulting manuscript is:

```
paper/degree3_spectral_refinement.pdf
```

4. Markdown-derived PDFs

With Pandoc, Python 3, and XeLaTeX available:

```
./build_markdown_pdfs.sh
```

The rendered files are written beneath `rendered/markdown/`, preserving the source directory structure. Their SHA-256 hashes are written to:

```
rendered/markdown/SHA256SUMS.txt
```

On macOS, verify the tracked files with:

```
shasum -a 256 -c rendered/markdown/SHA256SUMS.txt
```

On systems with GNU coreutils, use:

```
sha256sum -c rendered/markdown/SHA256SUMS.txt
```

5. Metadata and PDF preflight

Check consistency of the author name, AI-system identification, citation file, and licensing declaration:

```
./ci/preflight_metadata.sh
```

With Poppler installed, validate the manuscript and every Markdown-derived PDF:

```
./ci/preflight_pdfs.sh
```

The PDF check confirms that each PDF is nonempty and readable, has at least one page, contains extractable text, uses embedded fonts, and has a valid checksum when applicable. It also confirms that the manuscript contains the full author name, candidate coefficient, AI-assistance disclosure, and model designation.

What is exact and what is numerical

The general degree-3 tensor identities are checked with exact symbolic arithmetic by the Python/SymPy programs. The R programs are independently written numerical and finite-enumeration cross-checks. Agreement between the two languages helps detect implementation and convention errors, but it is not a substitute for mathematical review of the proof.

PDF reproducibility

The build scripts fix the source-date epoch and normalize PDF trailer identifiers. Repeated builds with an unchanged toolchain are intended to be byte-for-byte reproducible. Different operating systems, TeX Live releases, fonts, or Pandoc versions may still produce visually equivalent but byte-different PDFs. In the recorded 2026-09-04 clean-clone run, however, macOS with Pandoc 2.19.2 and TeX Live 2022 reproduced all tracked outputs byte for byte. The editable Markdown and LaTeX sources remain the source of truth.

GitHub Actions

The workflow runs automatically on pushes to `main`, pull requests, and manual requests from the Actions tab. It stores the Python and R logs and the generated PDFs as temporary workflow artifacts for inspection. A green run means that the repository's checks and builds completed in the declared environments; it is not an independent proof review.

Reporting a discrepancy

Record the following when reporting a failed or differing run:

- commit hash;
- operating system and architecture;
- Python, SymPy, NumPy, R, Pandoc, and TeX versions;
- the full command used; and
- the complete console output beginning at the first warning or error.